

Agenda

Short review of point estimators

Maximum likelihood estimators

Probability plots

Central Limit Theorem

Point estimators

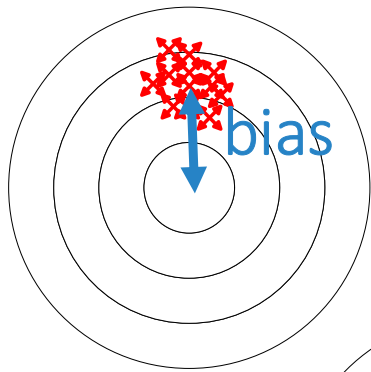
Let $X_1, \dots, X_n$ be a (simple) random sample with a common probability distribution $f_X(x; \theta)$, where θ is the parameter of the distribution.

Let $\hat{\Theta} = h(X_1, \dots, X_n)$, which is a statistic. $\hat{\Theta}$ is a random variable with a distribution that is a function of θ .

A statistic $\hat{\Theta}$ that is used to estimate the value of a parameter θ is called point estimator.

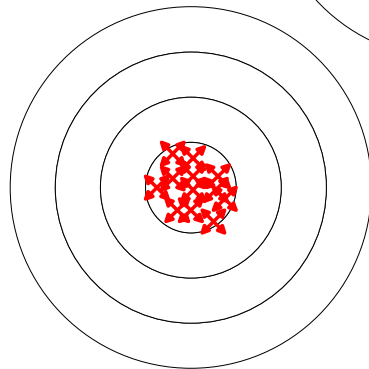
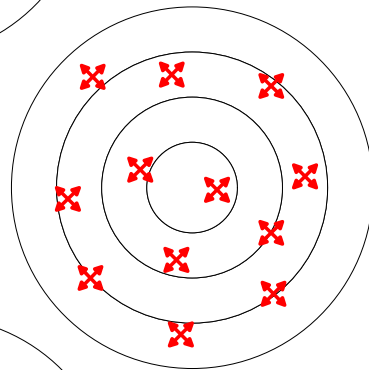
The distribution of $\hat{\Theta}$ depends on the distribution $f_X(x; \theta)$, so it depends on the parameter θ . Although θ is unknown, we can determine the distribution of $\hat{\Theta}$ or the expected value of $\hat{\Theta}$ or the variance of $\hat{\Theta}$ in terms of θ .

Mean Square Error of an Estimator



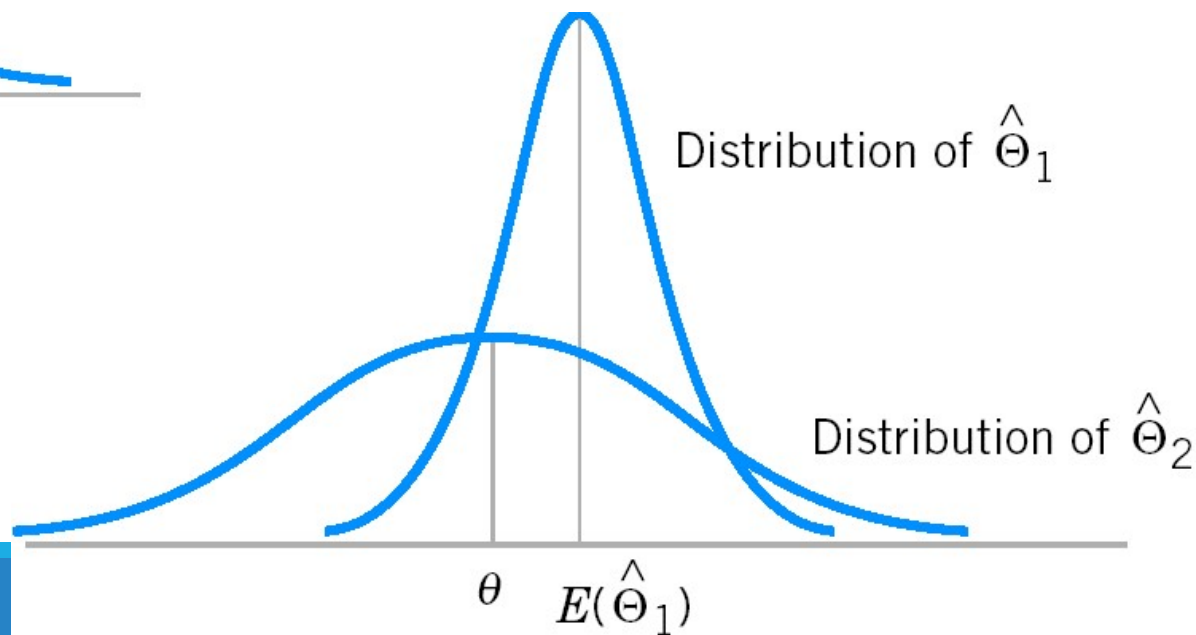
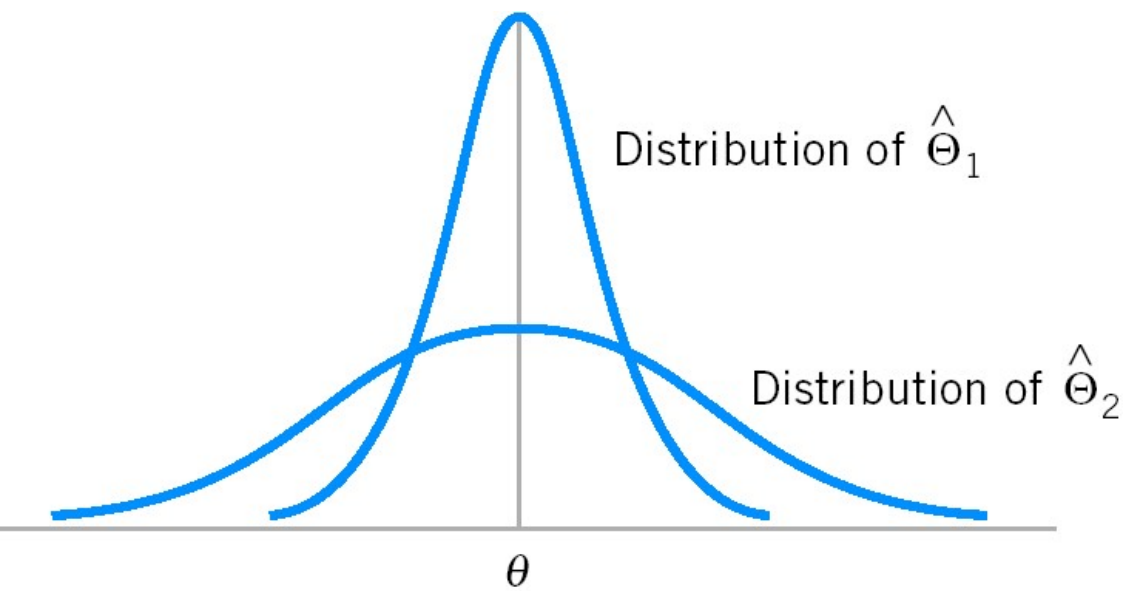
The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2$$



$$\text{MSE}_{\hat{\theta}} = \underbrace{(\mu_{\hat{\theta}} - \theta)^2}_{\text{Bias of } \hat{\theta}} + \underbrace{(\sigma_{\hat{\theta}})^2}_{\text{standard error of } \hat{\theta}}$$

General Concepts of Point Estimation



How to find good estimators?

Maximum likelihood estimators

Example:

A random sample of 10 new bike helmets manufactured by a certain company is obtained. Assume that we would like to estimate the probability $p = P\{\text{flawed helmet}\}$.

$$X_i = \begin{cases} 1 & \text{if helmet is flawed} \\ 0 & \text{otherwise} \end{cases}$$

Observed X_i 's = (1, 0, 1, 0, 0, 0, 0, 0, 0, 1)

What is the probability of observing this sequence in terms of p ,

i.e., $L(X_1, \dots, X_m; p) = ?$

Method of Maximum Likelihood

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \dots, x_n$ be the observed values in a random sample of size n . Then the **likelihood function** of the sample is

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \cdots \cdot f(x_n; \theta) \quad (7-5)$$

Note that the likelihood function is now a function of only the unknown parameter θ . The **maximum likelihood estimator** of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

Example:

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What is the probability of observing this sequence in terms of p , or the likelihood function $L(X_1, \dots, X_m; p) = ?$

Find $\hat{p}$ for which $L(X_1, \dots, X_m; p)$ is maximized.

Example

Sample: $X_1, \dots, X_n$, where X_i 's are normally distributed. Assume that σ^2 is known.

Find the maximum likelihood estimator of μ .

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Example

Sample: $X_1, \dots, X_n$, where X_i 's are normally distributed. **Assume that both parameters are unknown.**

Find the maximum likelihood estimators of μ and σ^2 .

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

A Desirable Property of the MLE

Proposition: Under very general conditions on the joint distribution of the sample, when the sample size n is large, the mle of any parameter θ :

- is approximately unbiased, i.e., $E(\hat{\theta}) \approx \theta$
- has a variance that is nearly as small as can be achieved by any estimator, i.e, the MLE is approximately the Minimum Variance Unbiased Estimator (MVUE) of θ .